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PARTICULARS OF THE EXPERIMENTS PERFORMED

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2.	The distribution of time ----- Fit negative exponential dist ⁿ . ----- Calculate expected frequencies				
3.	Let x has a gamma distribution ----- Find the parameter β and $P(x < 9.49)$				
4.	For the following Bivariate prob. dist ⁿ . ----- Find i) $P(x \leq 1, y = 1)$ ----- v) $P(x \geq 1, y \leq 2)$				
5.	Let (x, y) be a two continuous random variable with joint pdf $f(x, y) = K(2x + 3y)$ ----- Find -----				
6.	Joint prob. distribution ----- Find ----- iv) $F(2, 1), F(1, 1)$				

S.N.	Name of the experiment	Page No.	Date of experiment	Date of submission	Remarks
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8.	An IQ test was given to a random sample of 15 male and 20 female of students -----				
9.	The following are defective (D) and non-defective (N) - - - ----- Use Binomial test at the 1% level of significance				
10.	Out of 50 students willing ----- Use Binomial test to test the hypothesis -----				
11.	The data of percentage of broken groundnut pods - - ----- Test the hypothesis ----- Use sign test.				
12.	A survey of the time records ----- use sign test to test the hypothesis that the median -----				
13.	The weight (in lbs) at birth ----- Test the randomness of the 15 babies at 5% level of significance.				
14.	Two independent random samples ----- (i) ----- (ii) Test whether the average age of unemployed women is significantly high.				

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S.N.	Name of the experiment	Page No.	Date of experiment	Date of submission	Remarks
15.	The amount of rainfall (in cm) Do the data provide sufficient evidence to conclude				
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17.	The following are the number of misprints Use Kruskal wallis H test				
18.	The scores of 3 matched groups Apply the Friedman two way ANOVA test to see				
19.	The number of server manufactured using Wilcoxon matched pairs				
20.	Find laptop users Test whether there is difference				

S.N.	Name of the experiment	Page No.	Date of experiment	Date of Submission	Remarks
	between brands with respect to acceptability.				

Name: Aananda Kafle Statistics Practical Sheet

Date: .

Level: Bsc. 2nd year

Experiment No.: - 1

Shift: Day

Roll No.: 28

Group: A

OBJECT: Fit a negative binomial distribution to the following data

No. of emails per day (x)	0	1	2	3	4
No. of Days (f)	237	99	25	3	1

Also calculate the expected frequencies.

WORKING EXPRESSION

$$\mu_1' = \bar{x} = \frac{\sum fx}{N}, \mu_2' = \frac{\sum fx^2}{N}, \mu_2 = \sigma^2 = v(x) = \mu_2' - (\mu_1')^2$$

For negative Binomial distribution

$$\bar{x} = \frac{Kq}{p}, \sigma^2 = \frac{Kq}{p^2}$$

Recurrence relation, for Negative Binomial distribution

$$P(X=x+1) = \left(\frac{x+K}{x+1} \right) q p (X=x)$$

WORKING PROCESS

x	f	fx	fx ²
0	237	0	0
1	99	99	99
2	25	50	100
3	3	9	27
4	1	4	16
	N=365	$\sum fx = 161$	$\sum fx^2 = 242$

$$\mu_1' = \bar{x} = \frac{\sum fx}{N} = 0.4438$$

$$\mu_2' = \frac{\sum fx^2}{N} = 0.6630$$

$$\mu_2 = \sigma^2 = v(x) = \mu_2' - (\mu_1')^2 = 0.6630 - (0.4438)^2 = 0.46630$$

For NB(K, p)

$$\bar{x} = \frac{Kq}{p} \Rightarrow \frac{Kq}{p} = 0.4438 \quad \text{--- (1)}$$

$$\sigma^2 = \frac{Kq}{p^2} = 0.4660 \quad \text{--- (2)}$$

Dividing (1) by (2)

$$p = \frac{0.4438}{0.4660} = 0.9523 \quad \therefore q = 1 - p = 0.0477$$

Again,

$$\frac{Kq}{p} = 0.4438 \Rightarrow K = \frac{p \times 0.4438}{q} = \frac{0.9523 \times 0.4438}{0.0477}$$

$$\therefore K = 9$$

$$\text{Now, } P(X=0) = p^K = (0.9523)^9 = 0.6441$$

The other probability of NB are calculated by using recurrence relation.

$$P(X=x+1) = \left(\frac{x+K}{x+1} \right) q p (X=x)$$

Name : Aananda Kafle Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.:- 1

Shift : Day

Roll No. : 28

Group : A

x	$\frac{(2x+1)}{(x+1)} q$	$P(X=x)$	$E = N \cdot x \cdot P(X=x)$
0	$\frac{(0+1)}{(0+1)} \times 0.477 = 0.477$	$P(0) = 0.644$	$= 365 \times 0.644 \approx 235$
1	0.8385	$P(1) = 0.4283 P(0)$ $= 0.2765$	101
2	0.1749	$P(2) = 0.2385 \times P(1)$ $= 0.0659$	24
3	0.1431	$P(3) = 0.1749 \times P(2)$ $= 0.0115$	4
4	—	$P(4) = 0.1431 \times P(3)$ $= 0.0016$	1
			$E = 365$

CONCLUSION

Hence, the expected frequency is 365.

Name : Aaranda Kafle **Statistics Practical Sheet**

Date :

Level : Bsc 2nd year

Experiment No.:- 2

Shift : Day

Roll No. : 28

Group : A

OBJECT : The distribution of time (in second) to decay 100 identical radioactive particles is given below.

Decay Time	0-0.5	0.5-1.0	1.0-1.5	1.5-2.0	2.0-2.5
No. of particles	44	26	20	9	1

Fit a negative exponential distⁿ to the above data and also calculate expected frequencies.

WORKING EXPRESSION

$$\text{Mean } (X) = \frac{\sum fx}{N}, \quad P(a < X < b) = \int_a^b e^{-\theta x} \theta dx$$

For Negative exponential distribution

$$P(a < X < b) = \int_a^b \theta e^{-\theta x} dx$$

WORKING PROCESS

Let X be a r.v. that follow N.E.D. with parameter θ and the pdf is

$$f(x) = \theta e^{-\theta x}, \quad x > 0$$

Decay Time	No. of particles	mid value (X)	fx
0-0.5	44	0.25	11
0.5-1.0	26	0.75	19.5
1.0-1.5	20	1.25	25
1.5-2.0	9	1.75	15.75
2.0-2.5	1	2.25	2.25
	<u>$N=100$</u>		<u>$\sum fx = 73.5$</u>

$$\bar{x} = \frac{\sum fx}{N} = 0.735$$

$$\text{we have, } E(X) = \frac{1}{\theta} \Rightarrow \bar{x} = \frac{1}{\theta} \Rightarrow \theta = \frac{1}{\bar{x}}, \quad \theta = \frac{1}{0.735} = 1.36$$

Now,

$$\begin{aligned} P(0 < X < 0.5) &= \int_0^{0.5} \theta e^{-\theta x} dx = \int_0^{0.5} 1.36 e^{-1.36x} dx \\ &= [-e^{-1.36x}]_0^{0.5} = 1 - e^{-0.68} \\ &= 1 - 0.5066 \\ &= 0.4934 \end{aligned}$$

The remaining probabilities are computed in that manner

Fitting of N.E.D and calculation of expected frequency

Decay time (a-b)	$P(a < X < b)$	Expected freq = $NP(a < X < b)$
0-0.5	0.4934	49.34 $\approx$ 49
0.5-1.0	0.2500	25.0 $\approx$ 25
1.0-1.5	0.1266	12.66 $\approx$ 13
1.5-2.0	0.0642	6.42 $\approx$ 6
2.0-2.5	0.0325	3.25 $\approx$ 3
	<u>0.9667</u>	<u>97</u>

CONCLUSION

The expected frequency is 97.

Name : Aananda Kafle

Statistics Practical Sheet

Date :

Level : BSc 2nd year

Experiment No.:- 3

Shift : Day

Roll No. : 28

Group : A

OBJECT: Let X has a gamma distribution with pdf

$$f(x) = \frac{1}{\beta^2} x e^{-x/\beta} ; 0 < x < \infty$$

$$= 0 ; \text{elsewhere}$$

 If $x=2$ is the unique mode of the distribution.
 Find the parameter β and $P(X < 9.49)$

WORKING EXPRESSION

The mode of the gamma distribution is the solution of
 $f'(x) = 0$ and $f''(x) < 0$

CALCULATION

Here, $f(x) = \frac{1}{\beta^2} x e^{-x/\beta}$

and mode of the gamma distⁿ is $x=2$

That means, the first derivative of $f(x)$

$$f'(x) = \frac{1}{\beta^2} \frac{d}{dx} [x e^{-x/\beta}] = \frac{1}{\beta^2} \left[-\frac{x}{\beta} e^{-x/\beta} + e^{-x/\beta} \right]$$

$$\text{Making } f'(x) = 0, \frac{1}{\beta^2} \left[-\frac{x}{\beta} e^{-x/\beta} + e^{-x/\beta} \right] = 0$$

$$\Rightarrow \frac{e^{-x/\beta}}{\beta^2} \left[-\frac{x}{\beta} + 1 \right] = 0 \Rightarrow f(x) \left[-\frac{x}{\beta} + 1 \right] = 0$$

$x = \beta$ and we have given $x=2$

$$\therefore \boxed{\beta = 2}$$

Thus, $x \sim \text{Gamma}(2, \frac{1}{2})$ and its pdf is $f(x) = \frac{1}{4} x e^{-x/2}, x > 0$

$$\begin{aligned} P(X < 9.49) &= \int_0^{9.49} f(x) dx = \int_0^{9.49} \frac{1}{4} x e^{-x/2} dx \\ &= \frac{1}{4} \left[-\frac{1}{2} x e^{-x/2} + e^{-x/2} \right]_0^{9.49} \\ &= \frac{1}{4} \left[\left\{ -\frac{1}{2} \times 9.49 \times e^{-\frac{9.49}{2}} + e^{-\frac{9.49}{2}} \right\} + 1 \right] \\ &= \frac{1}{4} \times 0.97 = 0.235 \end{aligned}$$

RESULT

Hence, the parameter β of gamma distribution is 2
 and $P(X < 9.49)$ is 0.235.

Name : Aananda Kalle Statistics Practical Sheet

Level : Bsc. 2nd year

Experiment No.: - 4

Date :

Shift : Day

Roll No.: 28

Group : A

OBJECT: For the following Bivariate prob. distⁿ of X and Y.

X \ Y	0	1	2	3
0	0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
2	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Find (i) $P(X \leq 1, Y = 1)$
 (ii) $P(X < 0, Y \leq 2)$
 (iii) $P(X \leq 1)$
 (iv) $P(Y = 2)$ (v) $P(X \geq 1, Y \leq 2)$

WORKING EXPRESSION

Let (X, Y) be a two dimensional discrete random variable taking pair values (x_i, y_j) , $i = 1, 2, 3, \dots, m$ and $j = 1, 2, \dots, n$ then the prob function $P(X = x_i, Y = y_j)$ is called joint pmf of random variable (X, Y) if it satisfies the following conditions.

- $P(X = x_i, Y = y_j) \geq 0$
- $\sum_{i=1}^m \sum_{j=1}^n P(X = x_i, Y = y_j) = 1$

WORKING PROCESS

X \ Y	0	1	2	3	Total marginal pmf of X
0	0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$P(X=0) = \frac{1}{2}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$P(X=1) = \frac{1}{2}$
2	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$P(X=2) = \frac{1}{2}$
marginal	$P(Y=0) = \frac{1}{2}$	$P(Y=1) = \frac{1}{2}$	$P(Y=2) = \frac{1}{2}$	$P(Y=3) = \frac{1}{2}$	
Total Prob of Y = $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 2$					

$$i) P(X \leq 1, Y = 1) = P(X = 0, Y = 1) + P(X = 1, Y = 1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$ii) P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{1}{2} + \frac{1}{2} = 1$$

$$iii) P(X < 2, Y \leq 2) = P(X = 0, Y \leq 2) + P(X = 1, Y \leq 2) \\ = P(X = 0, Y = 0) + P(X = 0, Y = 1) + P(X = 0, Y = 2) + P(X = 1, Y = 0) + P(X = 1, Y = 1) + P(X = 1, Y = 2) \\ = 0 + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{2}$$

$$iv) P(Y = 2) = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$$v) P(X \geq 1, Y \leq 2) = P(X = 1, Y \leq 2) + P(X = 2, Y \leq 2) \\ = P(X = 1, Y = 0) + P(X = 1, Y = 1) + P(X = 1, Y = 2) \\ + P(X = 2, Y = 0) + P(X = 2, Y = 1) + P(X = 2, Y = 2) \\ = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{2}$$

RESULT

$$i) P(X \leq 1, Y = 1) = \frac{1}{2}$$

$$ii) P(X < 0, Y \leq 2) = 0$$

$$iii) P(X < 2, Y \leq 2) = \frac{3}{2}$$

$$iv) P(Y = 2) = \frac{3}{2}$$

$$v) P(X \geq 1, Y \leq 2) = \frac{3}{2}$$

Name : Ananda Kifle

Statistics Practical Sheet

Date :

Level : BSc. 2nd year

Experiment No.: - 5

Shift : Day

Roll No. : 28

Group : A

OBJECT : Let (X, Y) be a two dimensional continuous random variable with joint pdf: $f(x, y) = K(2x + 3y)$, $0 \leq x \leq 1$, $0 \leq y \leq 1$, otherwise $= 0$.

- Find (i) value of K
(ii) $P(X < 0.4, Y < 0.4)$
(iii) $P(X > 0.8, Y < 0.5)$

WORKING EXPRESSION

Let (X, Y) be a two dimensional continuous random variable then a continuous function $f(x, y)$ is said to be joint pdf of X, Y if it satisfies the following conditions.

(i) $f(x, y) \geq 0$ for all x and y .

(ii) $\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x, y) dx \right] dy = 1$

WORKING PROCESS

i) we have $\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x, y) dx \right] dy = 1 \Rightarrow \int_0^1 \left[\int_0^1 K(2x + 3y) dx \right] dy = 1$

$\Rightarrow \int_0^1 \left[2K \int_0^1 x dx + 3Ky \int_0^1 dx \right] dy = 1 \Rightarrow \int_0^1 \left[2Kx \cdot \frac{1}{2} + 3Ky \right] dy = 1$

$\Rightarrow K \int_0^1 dy + 3K \int_0^1 y dy = 1 \Rightarrow K(y)_0^1 + 3K \left(\frac{y^2}{2} \right)_0^1 = 1 \Rightarrow K + \frac{3}{2}K = 1$

$\Rightarrow \frac{5K}{2} = 1 \therefore K = \frac{2}{5}$

ii) $P(X < 0.4, Y < 0.4) = \int_0^{0.4} \left[\int_0^{0.4} f(x, y) dx \right] dy = \int_0^{0.4} \left[\frac{2}{5} \int_0^{0.4} (2x + 3y) dx \right] dy$
 $= \int_0^{0.4} \left[\frac{2}{5} \left(x^2 \right)_0^{0.4} + \frac{6}{5} y \left(x \right)_0^{0.4} \right] dy = \int_0^{0.4} \left[0.64 + \frac{6}{5} y \cdot 0.4 \right] dy$
 $= 0.64 \times 0.4 + \frac{0.48}{2} \times 0.4 = 0.256 + 0.096 = 0.352$

iii) $P(X > 0.8, Y < 0.5) = \int_0^{0.5} \left[\int_{0.8}^1 f(x, y) dx \right] dy = \int_0^{0.5} \left[\frac{2}{5} \int_{0.8}^1 (2x + 3y) dx \right] dy$
 $= \int_0^{0.5} \left[\frac{2}{5} \left(x^2 \right)_{0.8}^1 + \frac{6}{5} y \left(x \right)_{0.8}^1 \right] dy = \int_0^{0.5} \left[\frac{2}{5} \left(1 - 0.64 \right) + \frac{6}{5} y (1 - 0.8) \right] dy$
 $= \int_0^{0.5} \left[0.144 + 0.24y \right] dy = 0.144(y)_0^{0.5} + 0.24 \left(\frac{y^2}{2} \right)_0^{0.5}$
 $= 0.144 \times 0.5 + 0.24 \times \frac{0.5^2}{2} = 0.072 + 0.03 = 0.102$

RESULT

i) The value of K is $\frac{2}{5}$

ii) $P(X < 0.4, Y < 0.4) = 0.352$

iii) $P(X > 0.8, Y < 0.5) = 0.102$

Name: Ananda Kafle

Statistics Practical Sheet

Date :

Level: BSc 2nd year

Experiment No.: - 8

Shift : Day

Roll No. : 28

Group : A

OBJECT : Joint prob distribution of X and Y is

$Y \backslash X$	0	1	2
0	0.1	0.4	0.1
1	0.2	0.2	0

Find i) Prob distⁿ $P(X=x)$ of Xprob distⁿ $P(Y=y)$ of Y

ii) Are X and Y independent?

iii) the conditional prob distⁿ of Xgiven $Y=1$ iv) $F(2,1)$, $F(1,1)$

WORKING EXPRESSION

conditional prob mass function of X for given $Y=y$

$$P(X=x_i/Y=y_j) = \frac{P(X=x_i \cap Y=y_j)}{P(Y=y_j)}$$

Similarly

$$P(Y=y_j/X=x_i) = \frac{P(X=x_i \cap Y=y_j)}{P(X=x_i)}$$

The joint prob. distⁿ function, $F(x,y) = P(X \leq x, Y \leq y)$

$$= \sum_{j=0}^y \sum_{i=0}^x P(X=x_i, Y=y_j)$$

WORKING PROCESS

$$i) P(X=x) = \sum_{y=0}^1 P(X=x, Y=y)$$

$$P(X=x) = P(X=x, Y=0) + P(X=x, Y=1)$$

$$P(X=0) = P(X=0, Y=0) + P(X=0, Y=1) = 0.1 + 0.2 = 0.3$$

$$P(X=1) = P(X=1, Y=0) + P(X=1, Y=1) = 0.4 + 0.2 = 0.6$$

$$P(Y=y) = P(X=0, Y=y) + P(X=1, Y=y) + P(X=2, Y=y)$$

$$P(Y=0) = P(X=0, Y=0) + P(X=1, Y=0) + P(X=2, Y=0) = 0.1 + 0.4 + 0.1 = 0.6$$

$$P(Y=1) = P(X=0, Y=1) + P(X=1, Y=1) + P(X=2, Y=1) = 0.2 + 0.2 + 0 = 0.4$$

$$ii) \text{ Let } P(X=0) = 0.3, P(Y=0) = 0.6 \text{ then, } P(X=0) \times P(Y=0) = 0.18$$

$$P(X=0, Y=0) = 0.1$$

$$\therefore P(X=x, Y=y) \neq P(X=x) \times P(Y=y)$$

 $\therefore$ X and Y are dependent.

$$iii) \text{ The conditional prob distⁿ of X given } Y=1, P(X=x, Y=1) = ?$$

$$\text{we know, } P(X=x/Y=1) = \frac{P(X=x \cap Y=1)}{P(Y=1)}$$

$$P(X=0/Y=1) = \frac{P(X=0, Y=1)}{P(Y=1)} = \frac{0.2}{0.4} = 0.5$$

$$P(X=1/Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{0.2}{0.4} = 0.5$$

$$P(X=2/Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{0}{0.4} = 0$$

Name : Ananda Kalle

Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.:- 6

Shift : Day

Roll No. : 28

Group : A

$$iv) F(2,1) = P(X \leq 2, Y \leq 1)$$

$$\begin{aligned} &= P(X=0, Y \leq 1) + P(X=1, Y \leq 1) + P(X=2, Y \leq 1) \\ &= P(X=0, Y=0) + P(X=0, Y=1) + P(X=1, Y=0) + P(X=1, Y=1) + \\ &\quad P(X=2, Y=0) + P(X=2, Y=1) \\ &= 0.1 + 0.2 + 0.4 + 0.2 + 0.1 + 0 = 1 \end{aligned}$$

$$\begin{aligned} F(1,1) &= P(X \leq 1, Y \leq 1) = P(X=0, Y \leq 1) + P(X=1, Y \leq 1) \\ &= P(X=0, Y=0) + P(X=0, Y=1) + \\ &\quad P(X=1, Y=0) + P(X=1, Y=1) \\ &= 0.1 + 0.2 + 0.4 + 0.2 \\ &= 0.9 \end{aligned}$$

RESULT

i) $P(X=0) = 0.3$, $P(X=1) = 0.6$, $P(Y=0) = 0.6$, $P(Y=1) = 0.4$

ii) X and Y are dependent.

iii) The conditional prob. distⁿ of X given Y=1

$$P(X=0/Y=1) = 0.5$$

$$P(X=1/Y=1) = 0.5$$

$$P(X=2/Y=1) = 0$$

iv) $F(2,1) = 1$

$F(1,1) = 0.9$

Name : Ananda Kafle

Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.:- 7

Shift : Day

Roll No. : 28

Group : A

OBJECT: Two independent samples are given below

Sample 1: 10 11 8 8 14

Sample 2: 8 12 13 9 15 9 17

Test whether the two samples have come from the same population with respect to their medians. Use median test at 5% level of significance.

WORKING EXPRESSION

Test statistics of median test for small sample size (n_1, k, n_2, k)

$$P(A=a) = \frac{{}^{n_1}C_a \cdot {}^{n_2}C_{k-a}}{{}^kC_k} \quad \text{where, } a=0,1,2,\dots,\min(n_1,k), k=\frac{n_1+n_2}{2}$$

WORKING PROCESS

Arrange the data: 8, 8, 8, 9, 9, 10, 11, 12, 13, 14, 15, 17, $n=12$

$M_d = 0$ value of $\left(\frac{n+1}{2}\right)^{\text{th}}$ item = 6.5th item = 10.5

Now, the number of observations in the first sample less than or equal to median (a) = 3

first sample size (n_1) = 5, Second sample size (n_2) = 7

$$k = \frac{n_1+n_2}{2} = \frac{5+7}{2} = 6$$

Let M_{d1} and M_{d2} be median of I popⁿ and II popⁿ resp.

1) Formulation of Hypothesis:

(a) $H_0: M_{d1} = M_{d2}$

(b) $H_1: M_{d1} \neq M_{d2}$ (TTT)

2) Test statistics

$$P_0 = P(A=a) = \frac{{}^{n_1}C_a \cdot {}^{n_2}C_{k-a}}{{}^kC_k}, a=0,1,2,3,4,5(k)$$

3) Critical value

$$P_0 = P(A \geq a) = P(A \geq 3) = \frac{\sum {}^5C_a \cdot {}^7C_{6-a}}{{}^{12}C_6}$$

$$= \frac{{}^5C_3 \cdot {}^7C_3 + {}^5C_4 \cdot {}^7C_2 + {}^5C_5 \cdot {}^7C_1}{{}^{12}C_6}$$

$$\therefore P_0 = 0.5$$

4) Decision: Here $2P_0 > \alpha$ i.e. $2 \times 0.5 > 0.05$

Accept H_0 , Reject H_1

Here, H_0 is accepted that means it is insignificant i.e. the medians of popⁿ of sample 1 and sample 2.

CONCLUSION

Two samples have come from same population w.r. to median.

Name : Aananda Kafle Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.:- 8

Shift : Day

Roll No. : 28

Group : A

OBJECT: An IQ test was given to a random sample of 15 male and 20 female students of a university. Their scores were recorded as follows

Male : 59, 66, 62, 81, 75, 73, 83, 68, 68, 42, 70, 60, 77, 76, 44, 72

Female : 63, 77, 65, 71, 74, 60, 76, 61, 67, 72, 64, 45, 55, 82, 46, 53, 68, 78, 50, 81

WORKING EXPRESSION

Test statistics of median test for large sample size ($n_1 > 10, n_2 > 10$)

$$S^2 = \frac{N(ad-bc)^2}{(a+c)(b+d)(a+b)(c+d)} \sim \chi^2_1$$

WORKING PROCESS

Here, no. of male (n_1) = 15, No. of female (n_2) = 20

$$n = n_1 + n_2 = 35$$

a (No. of obs. of male $\leq M_d$) = 7, b (No. of obs. females $\leq M_d$) = 12

c (No. of obs. of male $> M_d$) = 8, d (No. of obs. of female $> M_d$) = 8

	Sample I	Sample II	Total
No. of obs $\leq M_d$	7(a)	12(b)	19 = a+b
No. of obs $> M_d$	8(c)	8(d)	16 = c+d
	a+c = 15	b+d = 20	N = 35

Let M_{d1} and M_{d2} be median IQ of male and female resp.

1. Formulation of Hypothesis: a) $H_0: M_{d1} = M_{d2}$
b) $H_1: M_{d1} \neq M_{d2}$

2. Test statistics:-
$$S^2 = \frac{N(ad-bc)^2}{(a+c)(b+d)(a+b)(c+d)} = \frac{35(7 \times 8 - 12 \times 8)^2}{15 \times 20 \times 19 \times 16} = 0.16$$

3. Critical value : $\chi^2_{(1)} = 3.84$

4. Decision : $\chi^2_{cal} < \chi^2_{tab}$, Accept H_0 , Reject H_1 at $\alpha = 5\%$

CONCLUSION

IQ of male and female are same in the university

Name : Ananda Kafle

Statistics Practical Sheet

Date :

Level : BSc. 2nd year

Experiment No.:- 9

Shift : Day

Roll No. : 28

Group : A

OBJECT : The following are defective (D) and non-defective (N) electronic items produced in given order by a certain machine
 N N D D N D N N N D D N D D N N D N D N N N
 Test whether defective and non-defective are equally produced or not.
 Use Binomial test at the 1% level of significance.

WORKING EXPRESSION

For small sample size ($n \leq 25$) of Binomial test

i) Test statistic : $x_0 = \min(n_1, n_2)$

ii) critical value : $P = P(X \leq x_0) = \sum_{x=0}^{x_0} {}^n C_x P^x q^{n-x} = \sum_{x=0}^{x_0} {}^n C_x \left(\frac{1}{2}\right)^n$

WORKING PROCESS

Here, number of N (n_1) = 13

number of D (n_2) = 9 (< 25)

1) Formulation of Hypothesis

① $H_0 : P = \frac{1}{2}$ ② $H_1 : P \neq \frac{1}{2}$ (TTT)

2) Test statistics

$$x_0 = \min(n_1, n_2) = \min\{13, 9\} = 9$$

3) critical value

$$P = \sum_{x=0}^{x_0} {}^n C_x \left(\frac{1}{2}\right)^n = \sum_{x=0}^9 {}^n C_x \left(\frac{1}{2}\right)^n = 0.262$$

$$P_0 = 2P = 2 \times 0.262, \text{ for T-T} \\ = 0.524 = 52.4\%$$

4) Decision :-

Here, $P_0 > \alpha\%$, where $\alpha = 1\%$

Accept H_0 , Reject H_1

CONCLUSION

Defective and non-defective are produced equally.

Name : Aananda Kifle

Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.:- 10

Shift : Day

Roll No. : 28

Group : A

OBJECT : Out of 50 students willing to express opinion of laptop 30 expressed preference to brand dell and 20 for brand lenovo. Use Binomial test to test the hypothesis that both brand of laptop are equally popular.

WORKING EXPRESSION

For large sample size ($n > 25$) of Binomial test

Test statistic : $x_0 = \min(n_1, n_2) = \min(20, 30)$

$$Z = \frac{(x_0 \pm 0.5) - np}{\sqrt{npq}} \quad \begin{array}{l} \text{use } +0.5 \text{ if } x_0 < np \\ \text{use } -0.5 \text{ if } x_0 > np \end{array}$$

WORKING PROCESS

Here, no. of students prefer dell laptop (n_1) = 30
no. of students prefer lenovo laptop (n_2) = 20

1) Formulation of Hypothesis

a) $H_0: P = \frac{1}{2}$

b) $H_1: P \neq \frac{1}{2} \text{ (TTT)}$

2) Test statistic

$$x_0 = \min(n_1, n_2) = \min(30, 20), \quad x_0 = 20$$

$$np = 50 \times \frac{1}{2} = 25, > x_0 \quad \therefore +0.5$$

$$Z = \frac{(x_0 + 0.5) - np}{\sqrt{npq}} = \frac{-0.27}{1} = -0.27$$

3) critical value : $|Z_{\alpha/2}|_{\text{Tab}} = 1.96$

4) Decision : $|Z|_{\text{cal}} < |Z|_{\text{tab}} = 1.96$

Reject H_1 and Accept H_0

CONCLUSION

Both brand of laptop are equally popular.

Name : Ananda Kalle

Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.: - 11

Shift : Day

Roll No. : 28

Group : A

OBJECT :- The data of percentage of broken groundnut pods recorded from an experiment are given below:-
9.8, 10.4, 11.5, 10.4, 8.5, 8.0, 10.5, 9.5, 8.8, 9.2
Test the hypothesis that the percentage of broken pods has the median 8. Use sign test.

WORKING EXPRESSION

For small sample size ($n \leq 25$) of sign test

i) Test statistic: Under H_0 ,

$$K = \min\{n(+), n(-)\}$$

ii) critical region: obtain P-value (P_0) from Binomial Table

$$P_0 = \sum_{y=0}^K n C_y p^y q^{n-y} = \left(\frac{1}{2}\right)^n \sum_{y=0}^K n C_y$$

WORKING PROCESS

1) Formulation of Hypothesis

(a) $H_0: M_d = 8$, i.e. the percentage of broken pods has the median 8 (M_{d0})

(b) $H_1: M_d \neq 8$ (TTT)

2. Test statistic:

Assign plus (+) sign, minus (-) sign and zero (0) to the given observations:-

i.e. find diff $x_i - 8$ and write their signs as follows

1.8	2.4	3.5	2.4	0.5	0	2.5	-0.5	0.8	1.2
+	+	+	+	+	0	+	-	+	+

Here, $n(+)=8$, $n(-)=1$, number of ties (t) = 1

Effective sample size (n_e) = $n - t = 10 - 1 = 9$

Therefore, the test statistic under H_0 is $K = \min\{n(+), n(-)\}$
 $= \min\{8, 1\} = 1$

3. Critical value:-

The P-value (P_0) is obtained as

$$P_0 = P(X \leq 1) = \left(\frac{1}{2}\right)^9 \sum_{y=0}^1 9 C_y = 0.020$$

For two tailed test (TTT) $P_0 = 2 \times 0.020 = 0.04$
 $P_0 = 4\%$

4. Decision

since $2P_0 = 4\%$ is less than $\alpha = 5\%$. Therefore, we accept H_1 and reject H_0 .

CONCLUSION

Hence, we conclude that the percentage of broken pods has not median 8.

Name : Ananda Kalle

Statistics Practical Sheet

Date :

Level : Bsc 2nd year

Experiment No.:- 12

Shift : Day

Roll No. : 28

Group : A

OBJECT : A survey of the time records of 120 randomly selected employees was conducted and a plus (+) sign was recorded. If the employee was absent 15 days or more and a minus sign if the employee was absent less than 15 days. Total showed 80 minus and 40 plus sign use the sign test to test the hypothesis that the median number of days, M_d , that an assembly line employee was absent is at least 15 days.

WORKING EXPRESSION

For large sample size ($n > 25$) of sign test

$$\text{test statistic : } z = \frac{K - np}{\sqrt{npq}}$$

$$\text{corrected } z = \frac{(K \pm 0.5) - np}{\sqrt{npq}}, \quad \begin{matrix} np > K, + \text{ sign} \\ np < K, - \text{ sign} \end{matrix}$$

WORKING PROCESS

Given, sample size (n) = 120 $n > 25$
 Total number of plus sign = $n(+)$ = 40
 Total number of minus signs = $n(-)$ = 80

1. Formulation of Hypothesis

- (a) Null hypothesis H_0 : $M_d \geq 15$, i.e. the number of plus sign is at least as many as the number of minus sign.
- (b) Alternative hypothesis H_1 : $M_d < 15$ (LTT)

2. Test statistic

$$\text{under } H_0, z = \frac{(K \pm 0.5) - \frac{1}{2} n}{\frac{1}{2} \sqrt{n}} \sim N(0,1)$$

$$= \frac{K \pm 0.5 - np}{\sqrt{npq}} \quad \text{where } K = \text{total number of less frequent sign} = n(-) = 40$$

$$z = \frac{(40 + 0.5) - 120 \times \frac{1}{2}}{\sqrt{120 \times \frac{1}{2} \times \frac{1}{2}}} = -3.5$$

3. Critical value:

$$P_0 = P(z \leq -3.562) = 0.002$$

- 4. Decision: Since, $P_0 = 0.0002 < 0.05 = \alpha$
 Accept H_1 , reject H_0

CONCLUSION

We conclude that there are significantly more minus signs than plus signs i.e. the median that an assembly line employee was absent 13 less than the claimed 15 days.

Name : Aananda Kofle

Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.:- 13

Shift : Day

Roll No. : 28

Group : A

OBJECT: The weights (in lbs) at birth of 15 babies born in a maternity hospital are given below:
 6.2, 7.6, 7.1, 6.9, 7.5, 8.7, 4.6, 6.8, 7.2, 8.1, 5.0, 5.8, 8.9, 8.5. Test the randomness of the 15 babies at 5% level of significance.

1. Formulation of Hypothesis

- ① H_0 : The weights of the 15 babies are in random order.
 ② H_1 : The weights of the 15 babies are not in random order.

2. Test statistic:

Under H_0 , the test statistic is R , the number of runs.

Calculation of 'R': Given $n=15$ and $\alpha=5\%$.

Arranging the weights in ascending order

4.6, 5.0, 5.8, 5.9, 6.2, 6.8, 6.9, 7.1, 7.2, 7.5, 7.6, 8.1, 8.5, 8.7, 8.9

The median M_d is given by

$$M_d = \text{Size of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ item} = \text{Size of } 8^{\text{th}} \text{ item}$$

$$M_d = 7.1$$

Now, Assign A to the given weights if $w_i > M_d$ and B to the given weights if $w_i < M_d$.

Then, we obtain the sequence on original data

BA B A BBB AAA BB AA

$$\therefore R=8$$

Here, since $w_3 = M_d$, a tie ($t=1$) is occurred the effective sample size

$$n_e = n - t = 15 - 1 = 14$$

3. Critical value:

for $n_1=7$ and $n_2=7$, the critical values x and $\bar{x}$ obtained from the table of run test at $\alpha=5\%$ level of significance are $x=3$ and $\bar{x}=13$

4. Decision:

since $R=8$ lies in $(x, \bar{x}) = (3, 13)$, there is no evidence to reject the null hypothesis.

Therefore, we accept H_0 and reject H_1 at $\alpha=5\%$

5. Conclusion:

Hence the weights of the 15 new born babies are in random order.

Name: Aananda Kafle

Statistics Practical Sheet

Date :

Level: BSc 2nd year

Experiment No.: - 14

Shift : Day

Roll No. : 28

Group : A

OBJECT: Two independent random samples of unemployed men and women are drawn and the ages of the 4 unemployed women and 5 unemployed men are recorded as follows:

women: 60 63 36 44

Men : 53 39 22 33 24

- Do these data represent sufficient evidence to conclude that there is a diff. in the average of unemployed men and women?
- Test whether the average age of unemployed women is significantly high

WORKING EXPRESSION

Values defined in terms of R_1 and R_2 for each sample as

$$V_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 \quad \left\{ \text{When } n_1 \leq 20 \text{ and } n_2 \leq 20 \right\}$$

$$V_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

where, R_1 = sum of the rank of 1st sample
 R_2 = sum of the rank of 2nd sample

WORKING PROCESS

1. Formulation of Hypothesis

⑤ $H_0: M_d1 = M_d2$ i.e. there is no diff. betⁿ the average age of unemployed women and men

⑥ $H_1: M_d1 > M_d2$

2. Test statistic:

calculation for test statistic V_0 .

combined ages in ascending order	Rank	Ranks of 1 st sample	Ranks of 2 nd sample
22	1		1
24	2		2
33	3		3
36	4	4	
39	5		5
44	6	6	
53	7		7
60	8	8	
63	9	9	
		$R_1 = 27$	$R_2 = 18$

As $n_1 = 4$ and $n_2 = 5$

$$V_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 4 \times 5 + \frac{4(4+1)}{2} - 27 = 3$$

$$V_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 = 4 \times 5 + \frac{5(5+1)}{2} - 18 = 17$$

$$\therefore V_0 = \min \{V_1, V_2\} = 3$$

3. Critical value:-

The prob. associated with the value as small as the observed value $V_0 = 3$

for $n_1 = 4$ and $n_2 = 5$ is 0.0556

i.e. $P_0 = P(V \leq 3) = 0.0556$

Name : Aaranda Kalle

Statistics Practical Sheet

Date :

Level : BSc. 2nd year

Experiment No.:- 14

Shift : Day

Roll No. : 28

Group : A

4. Decision

i) since $2P_0 = 2 \times 0.0556 = 0.1112 = 11.12\% > 5\% = \alpha$,
the data do not present sufficient evidence to reject the
null hypothesis H_0 .

ii) since $P_0 = 0.0556 > 0.05 = \alpha$, Accept H_0 .

5. Conclusion:-

Hence, we conclude that there is no sufficient diff.
betⁿ average age of unemployed women and men.

RESULT

Hence we can conclude that the average age of
unemployed women and men are equal.

Name : Aananda Kafle Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.:- 15

Shift : Day

Roll No. : 28

Group : A

OBJECT:- The amount of rainfall (in cm) of Nepal's two places Nagarkot and Phulchoki as revealed by the records of meteorological division, during monsoon were as follows:-

Nagarkot : 15 10 14 18 20 8 17 22 12 9
Phulchoki : 12 7 11 19 23 4 8 3 6 10
13 6 9 12 7 10 21 4 9 6
11 16 14

Do the data provide sufficient evidence to conclude significant lower than that at Nagarkot.

WORKING EXPRESSION

For large sample case ($n_1, n_2 > 20$) of Mann-Whitney U test

Test Statistic :- The sampling distribution is approximately normal with mean

$$E(U) = \mu_U = \frac{n_1 n_2}{2}, \sigma_U^2 = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

Therefore test statistic is

$$Z = \frac{U - \mu_U}{\sigma_U} = \frac{U - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}} \sim N(0, 1)$$

WORKING PROCESS

1. Formulation of Hypothesis

(a) $H_0: M_d1 = M_d2$ i.e. the average amount of rainfall of Nagarkot and Phulchoki are equal

(b) $H_1: M_d2 < M_d1$ i.e. the average amount of rainfall in Phulchoki is significantly lower than that at Nagarkot (OTI)

2. Test statistics:-

combined amount of rainfall in ascending order	Rank	Rank for 1st sample	Rank for 2nd sample	combined amount of rainfall	Rank	Rank for 1st sample	Rank for 2nd sample
3	1		1	12	20		20
4	2.5		2.5	12	20		20
4	2.5		2.5	13	22		22
6	5		5	14	23.5	23.5	
6	5		5	14	23.5		23.5
7	5		5	15	25.5	25.5	
7	7.5		7.5	15	25.5	25.5	
7	7.5		7.5	16	27		27
8	9.5	9.5		17	28	28	
8	9.5		9.5	19	29		29
9	12	12		20	30	30	
9	12		12	21	31		31
9	12		12	22	32	32	
10	15	15		23	33		33
10	15		15			$R_1 = 221$	$R_2 = 340$
10	15		15				
11	17.5		17.5				
11	17.5		17.5				
12	20	20					

Name : Aananda Kalle Statistics Practical Sheet

Date :

Level : BSc 2nd year

Experiment No.:- 15

Shift : Day

Roll No. : 28

Group : A

$$\text{Now, } U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 10 \times 23 + \frac{10(10+1)}{2} - 221 = 64$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 = 10 \times 23 + \frac{23(23+1)}{2} - 340 = 166$$

$$U_0 = \min(U_1, U_2) = \min\{64, 166\} = 64$$

Since, $n_1 = 10$ and $n_2 = 23$. It is a large sample case and therefore the test statistic for mann-whitney U test is

$$Z = \frac{U_0 - \mu_U}{\sigma_U} = \frac{U_0 - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}}$$

Since the ties are occurred betⁿ sample observations the test statistic should be corrected by applying a correction to the standard deviations σ_U for the ties.

$$\text{corrected } \sigma_U = \sqrt{\frac{n_1 n_2}{n(n-1)} \left[\frac{n^3 - n}{12} - \sum_{i=1}^n T_i \right]}$$

where $T_i = \frac{t_i^3 - t_i}{12}$ and t_i = number of times the i^{th} rank is repeated.

Since, the rank 9.5 is repeated 2 times, $t_1 = 2$

Similarly, $t_2 = 3, t_3 = 3, t_4 = 3, t_5 = 2$

$$\therefore \sum_{i=1}^5 T_i = \sum_{i=1}^5 \frac{t_i^3 - t_i}{12} = \frac{2^3 - 2}{12} + \frac{3^3 - 3}{12} + \frac{3^3 - 3}{12} + \frac{3^3 - 3}{12} + \frac{2^3 - 2}{12} = 7$$

$$\therefore \text{corrected } \sigma_U = \sqrt{\frac{n_1 n_2}{n(n-1)} \left[\frac{n^3 - n}{12} - \sum_{i=1}^n T_i \right]} = \sqrt{\frac{10 \times 23}{33(33-1)} \left[\frac{33^3 - 33}{12} - 7 \right]} = 25.49$$

Therefore, the corrected test statistic is

$$\text{corrected } Z = \frac{U_0 - \frac{n_1 n_2}{2}}{\text{corrected } \sigma_U} = \frac{64 - \frac{10 \times 23}{2}}{25.49} = -2$$

3. Critical value : $Z_{\alpha} = Z_{0.05} = -1.645$ (LTT)

$$|Z_{0.05}|_{\text{tab}} = 1.645$$

4. Decision : $|Z|_{\text{cal}} = 2 > |Z|_{\text{tab}}$, Reject H_0 , Accept H_1

CONCLUSION

Hence the data provide sufficient evidence to conclude that average amount of rainfall of phulchoki is significantly lower than that of Nagarkot at 5% level of significance.

Name : Ananda Kafle

Statistics Practical Sheet

Date :

Level : BSc. 2nd year

Experiment No.:- 16

Shift : Day

Roll No. : 28

Group : A

OBJECT: Three samples are taken comprising of 120 university teachers, 150 doctors and 130 advocates. Each person chosen is asked to select one of the three categories that best represents his feeling towards a certain national policy. The categories are in favour of policy (F), against the policy (A) and diff. towards the policy (I). The results of the interview are given below:

Sample	Reaction			Total
	F	A	I	
University Teachers	80	30	10	120
Doctors	70	40	40	150
Advocates	50	50	20	130

On the basis of these data, can it be concluded that reactions or views of university teachers, doctors and advocates are homogeneous in so far as a national policy under discussion is concerned?

WORKING EXPRESSION

Under H_0 , test statistics of Chi-square test

$$\chi^2 = \sum_{i=1}^3 \sum_{j=1}^3 \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$\text{where, } E_{ij} = \frac{R_i \times C_j}{n}$$

WORKING PROCESS

1. Formulation of Hypothesis

- (a) $H_0: F_1(x) = F_2(x) = F_3(x)$ for all x , i.e. the three population of university teachers, doctors and advocates are identical.
 (b) $H_1: F_1(x) \neq F_2(x) \neq F_3(x)$

2. Test Statistics:-

Under H_0 , the test statistics is

$$\chi^2 = \sum_{i=1}^3 \sum_{j=1}^3 \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \text{ in } \chi^2_{(3-1 \times 3-1)} = \chi^2_{(4)}$$

$$\text{where, } E_{ij} = \frac{R_i \times C_j}{n}, \quad \begin{matrix} i=1, 2, 3 \\ j=1, 2, 3 \end{matrix}$$

Sample	Reactions			Row Total R_i
	F	A	I	
University teachers	80	30	10	120
Doctors	70	40	40	150
Advocates	50	50	20	130
Column Total (C_j)	200	120	80	$n=400$

$$E_{11} = \frac{R_1 \times C_1}{n} = \frac{120 \times 200}{400} = 60, \quad E_{12} = \frac{R_1 \times C_2}{n} = \frac{120 \times 120}{400} = 36, \quad E_{13} = \frac{R_1 \times C_3}{n} = \frac{120 \times 80}{400} = 24,$$

$$E_{21} = \frac{R_2 \times C_1}{n} = \frac{150 \times 200}{400} = 75, \quad E_{22} = \frac{R_2 \times C_2}{n} = \frac{150 \times 120}{400} = 45, \quad E_{23} = \frac{R_2 \times C_3}{n} = \frac{150 \times 80}{400} = 30,$$

$$E_{31} = \frac{R_3 \times C_1}{n} = \frac{130 \times 200}{400} = 65, \quad E_{32} = \frac{R_3 \times C_2}{n} = \frac{130 \times 120}{400} = 39, \quad E_{33} = \frac{R_3 \times C_3}{n} = \frac{130 \times 80}{400} = 26$$

Name : Aananda Kafle Statistics Practical Sheet

Date :

Level : Bsc 2nd year

Experiment No.:- 16

Shift : Day

Roll No. : 28

Group : A

Therefore,

$$S^2 = \frac{(80-60)^2}{60} + \frac{(30-36)^2}{36} + \frac{(10-24)^2}{24} + \frac{(70-75)^2}{75} + \frac{(10-45)^2}{45} + \frac{(40-30)^2}{30} \\ + \frac{(50-65)^2}{65} + \frac{(50-39)^2}{39} + \frac{(30-26)^2}{26}$$

$$= 6.67 + 1 + 8.17 + 0.33 + 0.55 + 3.33 + 3.46 + 3.10 + 0.61$$

$$\therefore S^2_{cal} = 27.22$$

3. Critical value:

The critical value of S^2 at $\alpha = 5\%$ for $(3-1)/(3-1) = 2 \times 2 = 4$ degree of freedom is 9.49 i.e. $S^2_{0.05(4)} = 9.49$

4. Decision

Since the computed S^2 value, 27.22 is greater than the sufficient value 9.49. Therefore we reject H_0 , accept H_1 .

CONCLUSION

All the professionals university teachers, doctors and advocates are not homogeneous in giving their reactions on the national policy.

Name : Aananda Kalle Statistics Practical Sheet

Date :

Level : BSc. 2nd year

Experiment No.: -17

Shift : Day

Roll No. : 28

Group : A

OBJECT: The following are the numbers of misprints counted on pages selected at random from three editions of a book.

Edition I	4	10	2	6	4	12
Edition II	8	5	13	8	8	10
Edition III	7	9	11	2	14	2

Use Kruskal Wallis H test at 5% level of significance to test the null hypothesis that the samples come from identical populations.

WORKING EXPRESSION

Test statistics of Kruskal Wallis H test

$$H = \frac{12}{n(n+1)} \sum \frac{R_i^2}{n_i} - 3(n+1) \\ 1 - \frac{\sum (t_i^3 - t_i)}{n^3 - n}$$

WORKING PROCESS

	No of misprints						R_i	R_i^2/n_i
Edition I	4	10	2	6	4	12		
Rank	3.5	13.5	1.5	6	3.5	16	= 44	322.66
Edition II	8	5	13	8	8	10		
Rank	10	5	17	10	10	13.5	= 65.5	715.04
Edition III	7	9	11	2	14	2		
Rank	7.5	12	10	1.5	18	7.5	= 61.5	630.37
								$\sum \frac{R_i^2}{n_i} = 1668.076$

sample size of Edition I (n_1) = 6, sample size of Edition II (n_2) = 6
sample size of Edition III (n_3) = 6

No. of times rank 1.5 is repeated (t_1) = 2, No. of times rank 10 is repeated
No. of times rank 3.5 is repeated (t_2) = 2, (t_3) = 3
No. of times rank 7.5 is repeated (t_4) = 2, No. of times rank 13.5 is repeated
(t_5) = 2

$$\therefore \sum (t_i^3 - t_i) = (2^3 - 2) + (2^3 - 2) + (3^3 - 3) + (2^3 - 2) = 48$$

R_1, R_2, R_3 be sum of ranks for edition I, II and III resp.
Let Md_1, Md_2 and Md_3 be median misprint of edition I, II and III resp.

1. Formation of Hypothesis

- $H_0: Md_1 = Md_2 = Md_3$ i.e. population are identical
- $H_1: Md_1 \neq Md_2 \neq Md_3$ i.e. population are not identical

2. Test statistics: $H = \frac{12}{n(n+1)} \sum \frac{R_i^2}{n_i} - 3(n+1) = \frac{12}{6(6+1)} \times 1668.076 - 3(6+1)$
$$1 - \frac{\sum (t_i^3 - t_i)}{n^3 - n} = 1 - \frac{48}{18^3 - 18} = 1.5407$$

3. Critical value:- critical value at 5% level of significance for 2 degree of freedom is $\chi^2_{0.05(2)} = 5.99$

4. Decision:- $H = 1.54 < \chi^2_{0.05(2)} = 5.99$, Accept H_0 , Reject H_1

CONCLUSION

The samples come from identical population.

Name : Aananda Kafle

Statistics Practical Sheet

Date :

Level : Bsc. 2nd year

Experiment No.: -18

Shift : Day

Roll No. : 28

Group : A

OBJECT : The scores of 3 matched groups under the six conditions are given below.

Group	I	II	III	IV	V	VI
A	9	4	1	7	5	8
B	6	5	2	8	5	7
C	9	1	2	6	5	7

Apply the Friedman two way ANOVA test to see if there is significantly difference in variation between any two of the three matched groups

WORKING EXPRESSION

Friedman test statistic

$$F_r = \frac{12}{nk(k+1)} \sum R_i^2 - 3n(k+1)$$

$$1 - \frac{\sum (t_i^3 - t_i)}{nk(k^2 - 1)}$$

WORKING PROCESS

1) Test of Hypothesis

- a) H_0 : $M_dA = M_dB = M_dC$, i.e. scores of 3 matched group is same.
 b) H_1 : $M_dA \neq M_dB \neq M_dC$, i.e. scores of 3 matched group is not same.

2) Test statistic

Calculation of R_i

Group	Ranks for each condition						R_i	R_i^2
	I	II	III	IV	V	VI		
A	2.5	2	1	2	2	3	12.5	156.25
B	1	3	2.5	3	2	1.5	13	169
C	2.5	1	2.5	1	2	1.5	10.5	110.25
Total								435.5

Here, $k=3$, $n=6$, $\sum R_i^2 = 435.5$

and t_i for $i=1, 2, 3$ and 4 are

$t_1=2$, $t_2=2$, $t_3=3$, $t_4=2$

$$F_r = \frac{12}{nk(k+1)} \sum R_i^2 - 3n(k+1)$$

$$1 - \frac{\sum (t_i^3 - t_i)}{nk(k^2 - 1)} = \frac{12}{6 \times 3(3+1)} \cdot 435.50 - 3 \times 6 \times 4$$

$$1 - \frac{(2^3 - 2) + (2^3 - 2) + (3^3 - 3) + (2^3 - 2)}{6 \times 3(3^2 - 1)}$$

= 0.82

3. Critical value :- The critical value for $k=3$ and $n=6$ obtained from the table is 0.956

i.e. $P_0 = P(F_r \text{ corrected} = 0.82) = 0.956$

4. Decision : Since $P_0 = 0.956$ which is greater than $\alpha = 0.05$, we accept H_0 .

CONCLUSION

Hence the variation between any two groups of the three groups is not significant.

Name : Aananda Kafle

Statistics Practical Sheet

Date :

Level : BSc. 2nd year

Experiment No.:- 19

Shift : Day

Roll No. : 28

Group : A

OBJECT: The number of server manufactured from each of two companies A and B was recorded daily for a period of 10 days with the following results

Days	1	2	3	4	5	6	7	8	9	10
A	172	165	206	184	174	142	190	169	161	200
B	201	179	159	192	177	170	182	179	169	210

Assuming both companies produced same daily output, test the hypothesis that there is no difference betⁿ median defective hypothesis that company B produced more defective than company A using Wilcoxon Matched pairs signed signed test.

WORKING EXPRESSION

Test statistics for Wilcoxon matched pairs signed test.

$$T = \min\{S(+), S(-)\} \quad \therefore \text{for small sample size } n \leq 25\}$$

If $T_{cal} \leq T_{tab}$, Reject H_0
otherwise accept H_0 .

WORKING PROCESS

Days	Company A (x_i)	Company B (y_i)	$d_i = x_i - y_i$	R_i
1	172	201	29	9
2	165	179	14	7
3	206	159	-47	-10
4	184	192	8	3
5	174	177	3	1
6	142	170	28	8
7	190	182	-8	-3
8	169	179	10	5.5
9	161	169	8	3
10	200	210	10	5.5

Here,

$$n=10, \text{ sum of ranks of } + \text{ sign } (S(+)) = 9+7+3+1+8+5.5+3+5.5$$

$$= 42$$

$$\text{sum of ranks of } - \text{ sign } (S(-)) = 10+3=13$$

Let M_dA and M_dB be median of company A and B respectively

1. Formulation of Hypothesis

a) $M_dA = M_dB$ i.e. there is no significant difference in median defect of company A and company B.

b) $M_dA < M_dB$

2. Test statistic: $T = \min\{S(+), S(-)\} = \min\{42, 13\} = 13$

3. Critical value: $T_{0.05(10)} = 8$

4. Decision: $T_{cal} = 13 > T_{0.05(10)} = 8$
 T_{tab}

CONCLUSION

There is no difference between median defectives for two companies A and B.

Name : Ananda Kafle

Statistics Practical Sheet

Date :

Level : BSc 2nd year

Experiment No.: - 20

Shift : Day

Roll No. : 28

Group : A

OBJECT: Find laptop users were asked for the acceptability four brands for daily use. The response of acceptability (A) and rejection (R) are given below.

users	Brands			
	Alpha	Beta	Gamma	Delta
H ₁	A	R	A	R
H ₂	R	A	A	R
H ₃	R	A	R	A
H ₄	A	R	R	R
H ₅	A	A	R	A

Test whether there is any significant difference between brands with respect to acceptability.

WORKING EXPRESSION

Test statistics: Cochran Q-test

$$Q = \frac{(K-1) [K \sum R_i^2 - (\sum R_i)^2]}{K \sum C_j - \sum C_j^2} \times \chi^2$$

where, sum all the y (positive) according to treatments to get R_i (row wise) and according to objects to get C_j (column wise), i = 1, 2, 3, ..., k and j = 1, 2, 3, ..., n

WORKING PROCESS

Brands	users					R	R ²
	H ₁	H ₂	H ₃	H ₄	H ₅		
Alpha	(A)	R	R	(A)	(A)	3	9
Beta	R	(A)	(A)	R	(A)	3	9
Gamma	(A)	(A)	R	R	R	2	4
Delta	R	R	(A)	R	(A)	2	4
						$\sum R_i =$	
L _j	2	2	2	1	3	$\sum C_j = 10$	$\sum R_i^2 = 26$
C _j ²	4	4	4	1	9	$\sum C_j^2 = 22$	

Here, Number of brands (K) = 4
Number of users (n) = 5

1. Formulation of Hypothesis

- H₀: There is no significant diff. betⁿ brands
- H₁: There is at least one significant diff. betⁿ brands

2. Test statistic

$$Q = \frac{(K-1) [K \sum R_i^2 - (\sum R_i)^2]}{K \sum C_j - \sum C_j^2} = \frac{(4-1) [4 \times 26 - (10)^2]}{4 \times 10 - 22} = 0.666$$

3. Critical value:

$\chi^2_{Tab 0.05(3)} = 7.81$

4. Decision: Here, $Q < \chi^2_{0.05(3)}$, Accept H₀

CONCLUSION

There is no significant difference betⁿ brands according to acceptability.